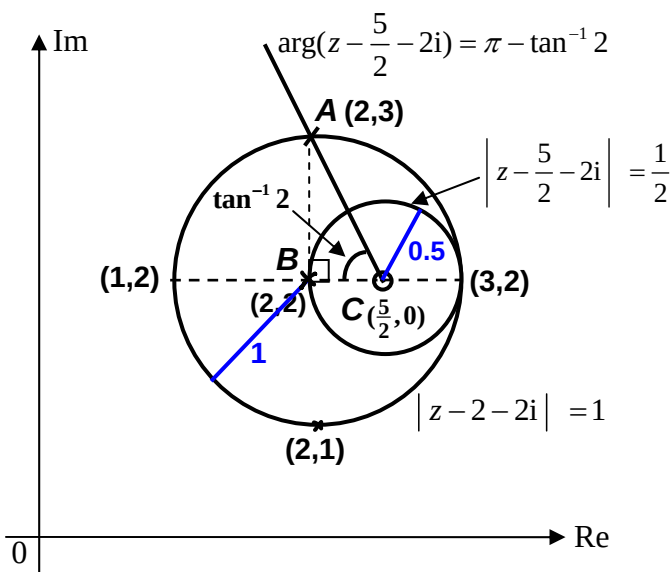
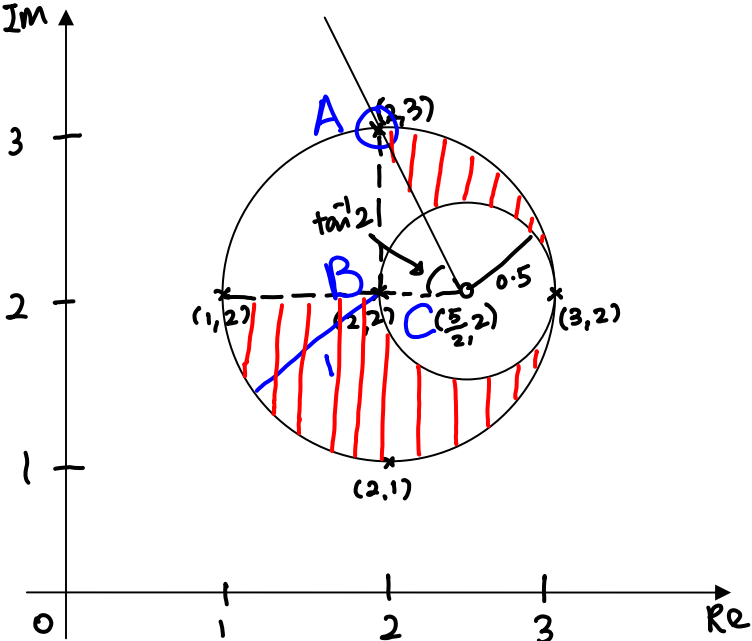
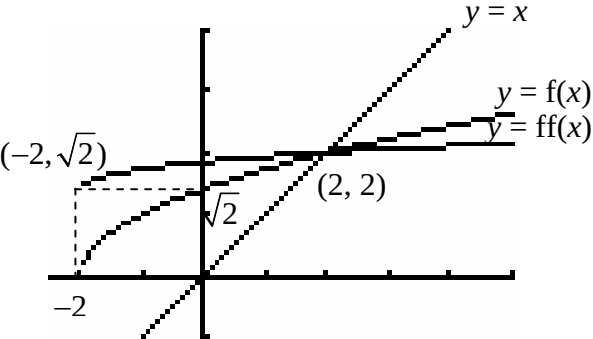


2009 RI(JC) H2 Maths Preliminary Examination Paper 2 (Solutions)

Section A: Pure Mathematics [40 marks]

1 (i) (ii) (iii)	
1(a)	Let point A be the point on the larger circle vertically above B where B be the centre of the larger circle and C be the centre of the smaller circle. Since radius of larger circle is 1, radius of smaller circle is 0.5, and the angle BCA is $\tan^{-1} 2$, BCA is a right angle triangle, and point A which is the intersection of the half line and the larger circle is at (2, 3). Hence $w = 2 + 3i$. Alternatively, Equation of half-line: $y = -2x + 7$ Equation of circle: $(x-2)^2 + (y-2)^2 = 1$. Substitution of one equation into another gives, $(x-2)^2 + (-2x+5)^2 = 1$ $x^2 - 4x + 4 + 4x^2 - 20x + 25 = 1$ $5x^2 - 24x + 28 = 0$ $(5x-14)(x-2) = 0$ $x = 2$ or $\frac{14}{5}$ (rej) Intersection is at (2, 3). Hence $w = 2 + 3i$.

1(b)	Shading as shown in diagram. 
2(i)	$R_f = [0, \infty) \subseteq [-2, \infty) = D_f$. Hence the composite function ff may be defined. $ff : x \mapsto \sqrt{\sqrt{x+2}+2}$, $x \geq -2$ and the range of ff is $[\sqrt{2}, \infty)$.
(ii)	
(iii)	$v = f(v)$. From the graph in part (b), the graph of $y = f(x)$ intersects the graph of $y = x$ at $x = 2$. Hence, $v = 2$.
(iv)	Range of $f^n = \{2\}$.
3(i)	Cost, $C = 0.09(2\pi r^2) + 0.06(2\pi rh)$ $= 0.18\pi r^2 + 0.12\pi r \left(\frac{300}{\pi r^2} \right)$ $= 0.18\pi r^2 + \frac{36}{r}$

	$\frac{dC}{dr} = 0.36\pi r - \frac{36}{r^2} = 0$ $r^3 = \frac{100}{\pi}$ $r = \sqrt[3]{\frac{100}{\pi}}$ $\frac{d^2C}{dr^2} = 0.36\pi + \frac{72}{r^3} > 0 \text{ when } r = \sqrt[3]{\frac{100}{\pi}}$ The cost is minimized when $r = \sqrt[3]{\frac{100}{\pi}}$. So $a = 100$
(ii)	Let V and H be the volume and height of the liquid respectively. When $t = 0$, $500\pi = \pi(2h)^2 h$ $h^3 = 125$ $h = 5$ $V = \pi(2h)^2 H$ $= 100\pi H$ $\frac{dV}{dt} = 100\pi \frac{dH}{dt}$ $= 100\pi(-0.02)$ $= -2\pi$ The volume of the liquid is decreasing at $2\pi \text{ cm}^3 \text{ s}^{-1}$ 2 minutes after the start of the experiment.
4(i)	A normal vector to plane is $\begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} \times \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ -6 \end{pmatrix}$ Equation of plane is $\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$ $\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix} = -5$
(ii)	$l_{AB}: \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}, \lambda \in \mathbb{R}$ Thus $\vec{OB} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$ for some λ.

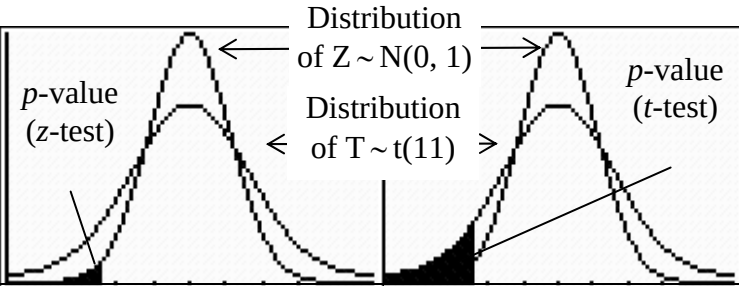
	Since B lies on the plane: $\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 1 \\ 6 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 6 \end{bmatrix} = -5 \Rightarrow 4\lambda + (2 + \lambda) + 36\lambda = -5$ $\Rightarrow \lambda = \frac{-7}{41}$ Thus $\vec{OB} = \frac{1}{41} \begin{pmatrix} -14 \\ 75 \\ -42 \end{pmatrix}$. $\vec{AB} = \frac{1}{41} \begin{pmatrix} -14 \\ 75 \\ -42 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \frac{1}{41} \begin{pmatrix} -14 \\ -7 \\ -42 \end{pmatrix}$ Perpendicular distance from A to $\Pi = \frac{1}{41} \sqrt{2009} = \frac{7\sqrt{41}}{41}$ units.
(iii)	Length of projection of $\vec{OA}$ on Π $= \left \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix} \right \times \frac{1}{\sqrt{41}} = \frac{1}{\sqrt{41}} \left \begin{pmatrix} 12 \\ 0 \\ -4 \end{pmatrix} \right = 4\sqrt{\frac{10}{41}}$
(iv)	Hence area of $\triangle OAB = \frac{1}{2} \left(\frac{7\sqrt{41}}{41} \right) \left(4\sqrt{\frac{10}{41}} \right)$ $= \frac{14\sqrt{10}}{41} \text{ units}^2$

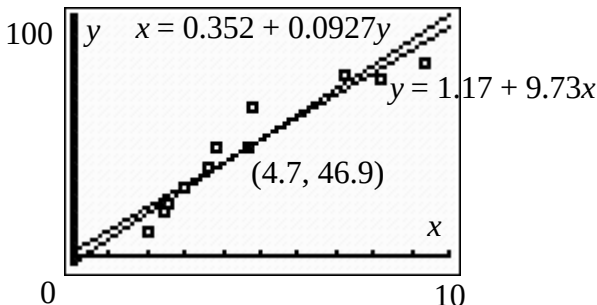
Section B: Statistics [60 marks]

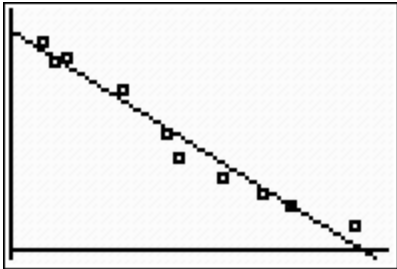
5(i)	Number of arrangements = $(4-1)! \times 4!$ = 144. $\square$
(ii)	Number of arrangements = $(7-1)! \times 2! \times 8$ = 11520. $\square$
6(a) (i)	Since A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$ $\Rightarrow P(B) = P(A \cup B) - P(A) = 0.6 - 0.2 = 0.4 \quad \square$
6(a) (ii)	Since A and B are independent, $P(A \cap B) = P(A) \times P(B) = 0.2 P(B) \dots (1)$ Also, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $\Rightarrow P(B) = P(A \cup B) - P(A) + P(A \cap B)$

	$\Rightarrow P(B) = 0.6 - 0.2 + 0.2P(B) \quad [\text{from (1)}]$ $\Rightarrow 0.8P(B) = 0.4 \Rightarrow P(B) = 0.5 \quad \square$
6(b) (i)	Required probability $= (0.1 \times 0.01) + (0.6 \times 0.15) + (0.3 \times 0.25)$ $= 0.166 \quad \square$
6(b) (ii)	$P(\text{good risks} \mid \text{no accidents}) = \frac{P(\text{good risks and no accidents})}{P(\text{no accidents})}$ $= \frac{0.1 \times (1 - 0.01)}{1 - 0.166} \quad [\text{from (i)}]$ $= \frac{0.099}{0.834} = \frac{33}{278} \text{ or } 0.119 \text{ (3 s.f.)}$
7(a)	Let Y be the number of observations of X that are less than 1, out of 88. Then $Y \sim B(88, 0.04)$. Since $n = 88 > 50$ is large and $p = 0.04 < 0.1$ is small, is such that $np = 3.52 < 5$, $Y \sim \text{Po}(3.52)$ approximately. Required probability $= P(Y \geq 6) = 1 - P(Y \leq 5)$ $= 1 - 0.85496 \text{ or } 0.14504 \text{ (5 s.f.)}$ $= 0.145 \text{ (3 s.f.)} \quad \square$
7(b) (i)	Let W be the number of larvae in 5ml of pond water. Then $W \sim \text{Po}(4 \times 0.5)$, i.e. $W \sim \text{Po}(2)$. $P(W < 3) = P(W \leq 2) = 0.67668 \text{ (5 s.f.)}$ $= 0.677 \text{ (3 s.f.)} \quad \square$
7(b) (ii)	Let T be the total number of larvae in 6 test-tubes, with each filled with 5ml of pond water. Then $T \sim \text{Po}(6 \times 2)$, i.e. $T \sim \text{Po}(12)$. Since $\lambda = 12 > 10$ is large, $T \sim N(12, 12)$ approximately. $P(T > 13) = P(T > 13.5) \quad (\text{by continuity correction})$ $= 0.33250 \text{ (5 s.f.)}$ $= 0.333 \text{ (3 s.f.)} \quad \square$
8 (i)	Given $X \sim N(20, \sigma^2)$. $P(12 < X < k) = P(X > k) = 0.42065$ $\Rightarrow P(X \leq 12) = 1 - 2(0.42065) = 0.1587.$ $\Rightarrow P\left(Z \leq \frac{12 - 20}{\sigma}\right) = 0.1587 \text{ where } Z \sim N(0, 1)$

	$\Rightarrow \frac{-8}{\sigma} = -0.99982 \text{ (5 s.f.) (from graphic calculator)}$ $\Rightarrow \sigma = \frac{-8}{-0.99982} = 8.0015 \text{ (5 s.f.)}$ Hence, $\sigma = 8$. (nearest whole number) (shown) $\square$ $P(X > k) = 0.42065$ $\Rightarrow P(X \leq k) = 1 - 0.42065 = 0.57935$ $\Rightarrow k = 21.6. \text{ (3 s.f.) (from graphic calculator) } \square$								
8 (ii)	Using $\sigma = 8$, $X \sim N(20, 8^2)$. $X = Y_1 + Y_2 - 4$ $\Rightarrow E(X) = E(Y_1 + Y_2 - 4)$ $\Rightarrow 20 = 2E(Y) - 4$ $\Rightarrow E(Y) = \frac{20+4}{2} = 12. \square$ Also, $\text{Var}(X) = \text{Var}(Y_1 + Y_2 - 4)$ $\Rightarrow 64 = 2 \text{Var}(Y) \text{ (since } Y_1 \text{ and } Y_2 \text{ are independent)}$ $\Rightarrow \text{Var}(Y) = 32. \square$ $\therefore Y \sim N\left(12, \left(\sqrt{32}\right)^2\right).$ $E(W - 2Y) = E(W) - 2E(Y) = 25 - 2(12) = 1.$ $\text{Var}(W - 2Y) = \text{Var}(W) + 2^2\text{Var}(Y) = 16 + 4(32) = 12^2.$ $(W - 2Y) \sim N(1, 12^2).$ $P(W > 2Y) = P(W - 2Y > 0) = 0.533. \text{ (3 s.f.) } \square$								
9 (i)	Select a sample of 50 students that is proportional to the size of each CCA type as shown in the table: <table><tr><td>Sports</td><td>Clubs</td><td>Performing Arts</td><td>Uniform Groups</td></tr><tr><td>$\frac{48}{150} \times 50 = 16$</td><td>$\frac{60}{150} \times 50 = 20$</td><td>$\frac{33}{150} \times 50 = 11$</td><td>$\frac{9}{150} \times 50 = 3$</td></tr></table> Within each stratum, use simple random sampling or systematic random sampling. $\square$	Sports	Clubs	Performing Arts	Uniform Groups	$\frac{48}{150} \times 50 = 16$	$\frac{60}{150} \times 50 = 20$	$\frac{33}{150} \times 50 = 11$	$\frac{9}{150} \times 50 = 3$
Sports	Clubs	Performing Arts	Uniform Groups						
$\frac{48}{150} \times 50 = 16$	$\frac{60}{150} \times 50 = 20$	$\frac{33}{150} \times 50 = 11$	$\frac{9}{150} \times 50 = 3$						
9 (ii)	Since the objective of the survey is to gather feedback on CCA experiences, using stratified random sampling ensures that all 4 CCA types are fairly represented in the sample, whereas simple random sampling does not ensure that. $\square$								

9 (iii)	Let $y = x - 5$. Then $\sum y = 32$ and $\sum y^2 = 400$. $\bar{y} = \frac{1}{n} \sum y = \frac{32}{50} = 0.64.$ $s_y^2 = \frac{1}{n-1} \left[\sum y^2 - \frac{1}{n} (\sum y)^2 \right] = \frac{1}{49} \left[400 - \frac{1}{50} (32)^2 \right]$ $= \frac{379.52}{49} = \frac{9488}{1225} = 7.7453. \quad (5 \text{ s.f.})$ Let $\bar{x}$ and s_x^2 be the unbiased estimates of the required mean and variance respectively. $\bar{x} = \bar{y} + 5 = 0.64 + 5 = 5.64. \quad \square$ $s_x^2 = s_y^2 = \frac{9488}{1225} = 7.75. \quad (3 \text{ s.f.}) \quad \square$
9 (iv)	Since $n = 50$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(5.64, \frac{7.7453}{50}\right) \text{ approximately.}$ i.e. $\bar{X} \sim N(5.64, 0.15491)$ approximately. $P(\bar{X} > 6) = 0.180. \quad (3 \text{ s.f.}) \quad \square$
10 (i)	Let X be the time taken to heat 2 litres of water from 25°C to 100°C, in seconds. Test: $H_0 : \mu = 440$ vs $H_1 : \mu < 440$, where $\mu = E(X)$. Perform a one-tail test at the 5% level of significance. Under H_0, $T = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} \sim t(n-1)$ where $\mu_0 = 440$, $\bar{x} = 428.92$, $s = 20.602$ and $n = 12$. Using a t-test, p-value = 0.0446 (3 s.f.) Since p-value = 0.0446 < 0.05, we reject H_0 and conclude that there is sufficient evidence, at the 5% level of significance, that the manufacturer's claim is valid. $\square$
10 (ii)	The z-test is the correct test because X follows a normal distribution and $\text{Var}(X)$ is known. $\square$
10 (iii)	Since the correct test (z-test) does not reject the null hypothesis, then p-value (for correct test) > 0.05. Also, from the graphs below, 

	we see that $p\text{-value (for } t\text{-test)} > p\text{-value (for } z\text{-test)}$ $\Rightarrow p\text{-value (incorrect test)} > p\text{-value (correct test)} > 0.05$ $\Rightarrow p\text{-value (incorrect test)} > 0.05$ Hence, we always do not reject the null hypothesis using the incorrect test, i.e. it is not possible for the incorrect test (t-test) to reject the null hypothesis. $\square$
11 (i)	From the graphic calculator, the least square regression line of y on x is 0 $y = 1.17088 + 9.7296x. \square$ the least square regression line of x on y is $x = 0.352 + 0.0927y. (3 \text{ s.f.}) \square$ and the product moment correlation coefficient for the data is 0.950. (3 s.f.) $\square$
11 (ii)	 From the graphic calculator, the scatter plot of y against x is as shown above. The required coordinates is $(\bar{x}, \bar{y})$, i.e. (4.7, 46.9). $\square$
11 (iii)	When $x = 12$, $y = 1.17088 + 9.7296(12) = 117.92608$. Hence, required predicted score = 117.92608 marks. $\square$ The estimate is not reliable as $x = 12$ is not within the domain of the given data set. $\square$ [OR the scatter plot in (ii) does not suggest a linear relationship between x and y.] $\square$ [OR the predicted score of $117.92608 > 100$ marks]
11 (iv)	From the graphic calculator, the least square regression line of y on $\frac{1}{x}$ is $y = 101 - \frac{198}{x}. (3 \text{ s.f.}) \square$ and when $x = 12$, $y = 101.498 - \frac{198.46}{12} = 85.0. (3 \text{ s.f.})$ Hence, required predicted score = 85.0 marks. (3 s.f.) $\square$

	The scatter plot of y against $\frac{1}{x}$ below suggests a linear relation between y and $\frac{1}{x}$, whereas that of y against x in (ii) does not. [OR as $x \rightarrow \infty$, $y \rightarrow \infty$ in (iii), but $y \rightarrow 101$ in (iv), which means that for large values of x, it is more realistic to use regression line in (iv) to estimate y.] Thus, this is a better estimate than that found in (iii). $\square$ 
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END